CHAPTER 8

ECOSYSTEM

CHAPTER ESSENCE AT A GLANCE:
Trophic productivity, nutrient decomposition, and unidirectional energy flow governed by thermodynamic laws in ecological webs.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Primary Productivity and Energy Capture:** Primary production is the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis. Gross Primary Productivity (GPP) is the total rate of organic matter production. Net Primary Productivity (NPP) is the available biomass for heterotrophic consumers after metabolic respiration losses: $NPP = GPP - R$. Annual net primary productivity of the whole biosphere is approximately 170 billion tons (oceans contribute only 55 billion tons despite occupying 70% of surface area).

• **The Five Steps of Decomposition:** Decomposition degrades complex organic detritus into inorganic carbon dioxide, water, and nutrients across five distinct phases: (1) Fragmentation (detritivores like earthworms break detritus into smaller particles), (2) Leaching (water-soluble inorganic nutrients percolate into soil horizons and precipitate as unavailable salts), (3) Catabolism (bacterial and fungal extracellular enzymes degrade detritus into simpler inorganic substances), (4) Humification (leads to accumulation of dark amorphous humus rich in nutrients and resistant to microbial action), and (5) Mineralization (microbes degrade humus to release inorganic minerals).

• **Lindeman Ten Percent Law of Energy Flow:** Energy flow is unidirectional from the sun through producers to consumers. Green plants capture only 2 to 10 percent of incident photosynthetically active radiation (PAR). Lindeman's 10% law dictates that only about 10 percent of energy is transferred from one trophic level to the next higher level, with 90 percent dissipated as metabolic heat into the environment conforming to the Second Law of Thermodynamics.

• **Ecological Pyramids and Inversions:** Pyramid of numbers is upright in grasslands but inverted in a single tree ecosystem supporting hundreds of birds and parasites. Pyramid of biomass is upright in forests but inverted in open marine ecosystems where small standing crops of short-lived phytoplankton support larger biomasses of zooplankton and predatory fish. The pyramid of energy is ALWAYS upright without exception, because energy is irreversibly lost as heat at every trophic step.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\mathrm{NPP} = \mathrm{GPP} - R \quad [\mathrm{GPP}: \text{ Gross Primary Production, } R: \text{Respiratory Energy Loss}]$$
Net Primary Productivity Formulation

$$\text{Energy Transfer} = \mathrm{Trophic}_{n+1} = 0.10 \times \mathrm{Trophic}_{n} \quad [\text{Lindeman } 10\% \text{ Ecological Law}]$$
Thermodynamic Trophic Transfer Efficiency

$$\text{Decomposition Cascade} = \text{Fragmentation} \rightarrow \text{Leaching} \rightarrow \text{Catabolism} \rightarrow \text{Humification} \rightarrow \text{Mineralization}$$
Five Sequential Enzymatic and Physical Decomposition Stages

$$\text{Marine Biomass Inversion} = \text{Phytoplankton } (4 \text{ g m}^{-2}) \longrightarrow \text{Zooplankton and Fish } (21 \text{ g m}^{-2})$$
Inverted Biomass Pyramid in Aquatic Marine Ecosystems

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Ecological Succession, Nutrient Cycling (Carbon and Phosphorus), and Ecosystem Services were completely deleted under the rationalized syllabus.

• **Alert:** The pyramid of energy is ALWAYS upright with zero exceptions; it can never be inverted because energy is lost as heat at each transfer.

• **Alert:** In open oceans, the pyramid of biomass is inverted because phytoplankton have microscopic biomass and high turnover rates, sustaining a much larger standing biomass of long-lived fish.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Gross Primary Productivity (GPP) and Net Primary Productivity (NPP). Write the mathematical relationship connecting them. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Gross Primary Productivity (GPP): The total rate of production of organic matter by green plants during photosynthesis per unit area and time. **[0.75 Mark]**

• Net Primary Productivity (NPP): The remaining biomass available to heterotrophic consumers (herbivores and decomposers) after respiratory utilization by plants. **[0.75 Mark]**

• Mathematical relationship: NPP = GPP - R (where R represents respiratory energy loss). **[0.5 Mark]**

2. Why is the pyramid of energy always upright with no exceptions in any ecosystem? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Energy flow is strictly unidirectional from primary producers to top carnivores. **[1 Mark]**

• In accordance with the Second Law of Thermodynamics, only about 10 percent of energy is conserved at each trophic transfer, while 90 percent is lost as metabolic heat, meaning higher trophic levels always receive less energy. **[1 Mark]**

3. Describe the five sequential steps involved in the decomposition of dead organic detritus in a terrestrial ecosystem. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Fragmentation: Detritivores (e.g., earthworms) physically break down dead plant and animal detritus into smaller particles. **[0.6 Mark]**

• Leaching: Water-soluble inorganic nutrients dissolve in percolating water and precipitate into deep soil horizons as unavailable salts. **[0.6 Mark]**

• Catabolism: Bacterial and fungal extracellular digestive enzymes degrade detritus into simpler inorganic substances. **[0.6 Mark]**

• Humification: Leads to the accumulation of dark, amorphous, colloidal humus that is highly resistant to microbial action and serves as a nutrient reservoir. **[0.6 Mark]**

• Mineralization: Humus is further slowly degraded by specialized microbes to release dissolved inorganic mineral nutrients into the soil. **[0.6 Mark]**

4. Explain Lindeman's 10 percent law of energy transfer across trophic levels. If 1,000,000 Joules of sunlight fall on green plants, calculate the energy available to a tertiary consumer. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Ten percent law: Only about 10 percent of the organic energy stored in a trophic level is transferred and incorporated into the biomass of the next higher trophic level. **[1 Mark]**

• Primary producers (Plants): Capture only 1% of incident solar energy = 1,000,000 * 0.01 = 10,000 Joules. **[0.5 Mark]**

• Primary consumers (Herbivores): 10% of 10,000 J = 1,000 Joules. **[0.5 Mark]**

• Secondary consumers (Carnivores): 10% of 1,000 J = 100 Joules. **[0.5 Mark]**

• Tertiary consumers (Top carnivores): 10% of 100 J = 10 Joules available. **[0.5 Mark]**

5. Explain why the pyramid of biomass in an aquatic (oceanic) ecosystem is inverted, whereas in a grassland ecosystem it is upright. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Aquatic inverted pyramid: Phytoplankton are microscopic primary producers with small standing crop biomass at any instant, but reproduce and turn over extremely rapidly. **[1.5 Marks]**

• They support a much larger standing biomass of long-lived zooplankton and large predatory fish, producing an inverted biomass profile. **[0.5 Mark]**

• Grassland upright pyramid: Grasses have enormous standing crop biomass, supporting smaller biomass of herbivores (rabbits, deer), which in turn support still smaller biomass of apex carnivores (foxes, lions), yielding an upright pyramid. **[1 Mark]**

6. Differentiate between a Grazing Food Chain (GFC) and a Detritus Food Chain (DFC) regarding their energy source, primary organisms, and ecological dominance. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Grazing Food Chain (GFC): Begins with living autotrophic green plants that capture solar energy; major pathway of energy flow in aquatic ecosystems (e.g., Phytoplankton -> Zooplankton -> Fish). **[1.5 Marks]**

• Detritus Food Chain (DFC): Begins with dead organic matter (detritus) broken down by saprophytic heterotrophic decomposers (fungi, bacteria); represents the dominant pathway of energy flow in terrestrial ecosystems. **[1.5 Marks]**

7. An industrial accident causes severe acid rain that destroys soil bacterial and fungal populations in a forest without killing mature trees. Explain the ecological consequences on decomposition, nutrient cycling, and humus formation. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Blockade of decomposition: Decomposers (bacteria and fungi) are the primary enzymatic agents of catabolism; without them, fallen leaves, branches, and dead animals cannot be broken down and will pile up indefinitely. **[1.5 Marks]**

• Arrest of humus formation: Humification ceases completely because amorphous colloidal humus is a product of partial microbial digestion of detritus. **[1 Mark]**

• Arrest of mineral recycling: Mineralization stops, so essential inorganic nutrients (nitrogen, phosphorus, potassium) remain locked within dead detritus rather than returning to the soil solution. **[1 Mark]**

• Ultimate forest collapse: Mature trees will eventually exhaust soil mineral reserves, leading to nutrient starvation and ecosystem collapse. **[0.5 Mark]**